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ORIGINAL ARTICLE

Effects of *Ginko biloba* leaf extract on the neurogenesis of the hippocampal dentate gyrus in the elderly mice

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Abstract Aging is associated with reduced hippocampal neurogenesis, which may in turn contribute to cognitive impairment. We assessed the effect of *Ginkgo biloba* (Gb) on hippocampal neurogenesis in elderly male mice using immunohistochemistry. We used anti-caspase-3 as a marker of apoptosis, anti-GFAP as a marker of neural stem cells, anti-Ki-67 as a specific marker for cellular proliferation and anti-doublecortin (DCX) to detect newly born neurons in the hippocampal dentate gyrus (DG) of aged male mice. The 24-month-old male mice were divided into two groups: a control group treated with distilled water and a group fed with Gb at a dose of 100 mg/kg once daily for 28 days. A sharp decrease in apoptotic cells in Gb-treated compared to nontreated mice was observed by anti-caspase-3 immunostaining. A large number of GFAP+ve cells was found in the subgranular zone of the DG of Gb-treated mice, suggesting an increase in the pool of neural stem cells by Gb treatment. There was also an increase in Ki-67 immunoreactive cells, indicating increased cell proliferation in the DG in the Gb-treated compared to nontreated group. A significant increase in newborn DCX+ve neurons with well-developed tertiary dendrites was also found in the Gb-treated compared to nontreated group. Using Western blot analysis, the expression of DCX protein in the Gb group was also significantly increased compared to the control. The results support a beneficial role of Gb on hippocampal neurogenesis in the context of brain aging.

Keywords *Ginkgo biloba* leaf extract · Immunohistochemistry · Neurogenesis · Old-aged mice

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Introduction

Previous studies have shown that several regions in the brain have the ability to generate new neurons and glia. Among the neurogenic areas is the hippocampal dentate gyrus (DG), which is a very important region. In the DG of the hippocampus, the neural progenitor cells, which reside in the subgranular zone (SGZ) of the DG, can divide and migrate into the granular cell layer and finally participate in the hippocampal neural circuity (Dayer et al. 2003; Kee et al. 2007; Toni et al. 2007). Also the hippocampus is susceptible to diverse neurological diseases such as Alzheimer's and ischemia (Tang et al. 2002; Stackman et al. 2003; Brinton and Wang 2006; Hwang et al. 2010).

The age-dependent reduction in adult neurogenesis (Kuhn et al. 1996; Encinas and Sierra 2012; Gebara et al. 2013) is associated with decreased learning performance (Gil-Mohapel et al. 2013), which can be restored by increasing adult neurogenesis with natural agents, voluntary exercise or drugs (van Praag et al. 2005). Thus, manipulations aimed at increasing adult neurogenesis represent a promising approach to alleviating age-related mood disorders (Drew and Hen 2007) as well as cognitive impairment (Bolognin et al. 2014).

Ginkgo biloba (Gb) leaf extract is known for its specific actions in improving blood flow, protective actions against damage by free radicals and antiinflammatory effects (Kotakadi et al. 2008; Tada et al. 2008; Wu et al. 2008; Shi et al. 2009). In the brain, one of the important actions of Gb is the positive effect on learning, concentration and memory (O'Hara et al. 1998; Diamond et al. 2000; Shif et al. 2006). Ginkgo also enhances working memory (Rigney et al., 1999; Wesnes et al. 2000), learning memory (Winter 1998) and long-term memory (Wesnes et al. 2000). However, these studies have not addressed whether Gb increases net hippocampal neurogenesis in vivo or not.

This study assessed the effect of Gb on the age-related decline of adult hippocampal neurogenesis. To this aim, we tested the effect of Gb leaf extract administration on elderly mice that were 2 years old, an age at which adult neurogenesis has reached its minimal activity (Gil-Mohapel et al. 2013), using the immunohistochemical identification of adult DG hippocampal stem cells and neurogenesis. The neural stem cells are slowly proliferating astrocytes and are identified on the basis of their expression of glial fibrillary acidic protein (GFAP). Therefore, in this study, GFAP is used as a neural stem cell marker in astrocytes residing in the subventricular zone of the DG (Von Bohlen und Halbach 2007). Ki-67 is expressed during mitosis in mammalian species (Scholzen and Gerdes 2000; Kee et al. 2002). Therefore, Ki-67 is used as an endogenous marker of cell proliferation. In addition, doublecortin (DCX), one of the microtubule-associated phosphoproteins, is expressed in the migratory neuroblasts and newly generated neurons (Francis et al. 1999; Gleeson et al. 1999). We examined the effect of Gb leaf extract on the main steps of the formation of new neurons in the aging Gb-treated hippocampus.

In this study, we adapted oral administration of Gb because in traditional medicine Gb has always been taken orally. In the present study, administered Gb may have crossed the blood-brain barrier because a single intravenous administration of Gb (8 mg/kg) resulted in a significant level of Gb in both the plasma and brain of rats (Madgula et al. 2010). We also adjusted the dose to 100 mg/kg body weight because previous studies showed the beneficial effect of this dose on hippocampal neurogenesis (Shif et al. 2006; Yoo et al. 2011).

Materials and methods

Ethics statement

Handling and care of animals conformed to the guidelines, being in compliance with current national and international laws and policies (NIH Guide for the Care and Use of Laboratory Animals, NIH publication no. 85–23, 1985, revised 1996). All of the experiments were conducted to minimize the number of animals used and suffering caused by the procedures used in the present study.

Animals and Gb leaf extract administration

Thirty male mice (24 months old) and ten male mice (young adults, 3 months old) were used in this study. The animals were housed in a conventional state at adequate temperature (22 °C) with humidity (55 %) control with a 12-h light/12-h dark cycle and had free access to food and tap water. The elderly mice were divided into two equal groups: the control distilled water (DW)-treated group and Gb-treated group. The treated group was given a dose of 100 mg/kg (according to Yoo et al. 2011). Since DCX is exclusively expressed in immature neurons from 1 to 28 days of cell age (Brown et al. 2003; Couillard-Despres et al. 2005), the animals were orally fed with DW or Gb using a feeding needle once daily for 28 days and were killed 2 h after the last administration. The animals of both groups were anesthetized with 30 mg/kg chloral hydrate (Sigma, St. Louis, MO, USA) i.p. then perfused transcardially with 0.1 M phosphate-buffered saline (PBS, pH 7.4) followed by 4 % paraformaldehyde in 0.1 M phosphate buffer (PB, pH 7.4).

Hematoxylin and eosin (H&E) staining

Five animal brains were used from each group (control, i.e., the elderly nontreated group and elderly Gb-treated group) with an additional five nontreated young adult mice brains used for comparison. All were stained with H&E. In summary, the fixed brains were dehydrated in different grades of ethanol: 70, 80, 90 and finally 100 %. Then the whole brains were mounted with hot paraffin and left to become solid to form paraffin blocks. Coronal sections were made, and slides (5 µm thick) containing paraffin were placed in a slide holder (glass or metal) and deparaffinized, then mounted in xylene. The slides were rehydrated in different grades of 100 % ethanol, 95 % ethanol and 80 % ethanol. Then they were mounted in deionized H_2O. Hematoxylin staining was done using hematoxylin dye (Poly-Scientific, Bayshore, NY, #s212A; Harris hematoxylin with glacial acetic acid), rinsed with deionized water, then tap water, dipped in 12× (fast) acid ethanol (to destain) and rinsed in tap water. Eosin staining was done by mounting 30 s in eosin (Poly-Scientific, Bayshore, NY; #s176; eosin phloxine stain,), then dehydrating by 95 % ethanol and 100 % ethanol and finally mounting in xylene Then the slides were covered using Permount (Fisher Scientific #SP15-100; histological mounting medium).

Immunohistochemistry

Five fixed brains were used in each group for this study (control elderly and Gb-treated elderly mice groups). Five additional normal young adult male mice were used for anti-caspase-3 staining to check for apoptosis in this age group for comparison. The 30-μm-thick brain sections in coronal plane were serially cut using a cryostat (Leica, Wetzlar, Germany). Slices were kept in cryoprotectant (30 % ethylene glycol and 25 % glycerin in 1× PBS) at −20 °C until being processed for immunostaining. Tissue sections were selected between 1.46 mm (the point of the first section taken) and 2.46 mm (the point of the last section taken) posterior to the bregma in reference to the mouse atlas (Franklin and Paxinos 1997) for each animal. The sections were sequentially treated with 0.3 % hydrogen peroxide (H_2O_2) in PBS for 30 min and 10 % normal goat or rabbit serum in 0.05 M PBS for 30 min. They were next incubated with diluted rabbit anti-caspase-3, anti-GFAP or anti-Ki-67 (1:1,000, Abcam, Cambridge, UK) or goat anti-DCX antibody (1:50, Santa Cruz Biotechnology, Santa Cruz, CA, USA) overnight at room temperature and subsequently exposed to biotinylated rabbit anti-goat or goat anti-rabbit IgG (diluted 1:200, Vector, Burlingame, CA, USA) and streptavidin peroxidase complex (diluted 1:200, Vector). Then, the sections were visualized with reaction to 3,3′-diaminobenzidine tetrachloride (Sigma) in 0.1 M Tris-HCl buffer (pH 7.2), counterstained with hematoxylin and mounted on gelatin-coated slides.

Counting cells

All images were acquired using a light microscope equipped with a digital camera. The total numbers of immunoreactive cells throughout the entire granule cell layer were estimated using stereological sampling, as previously described (Thuret et al. 2009). All cells were counted blind with regard to the mouse status. The total number of immunolabeled cells were counted in the entire thickness of the sections in a 1-in-6 series (240 μm apart) with a 40× objective. Cells expressing Ki-67, DCX and GFAP with a prototypical stellar astrocyte morphology were counted in the granule cell layer and subgranular zone.

The dendritic complexity of DCX-positive cells was traced and counted. DCX-positive cells were separated into three categories according to dendritic complexity: (1) lacking dendrites, (2) having immature dendrites (primary or secondary branches that do not extend into the outer molecular layer) and (3) having mature dendrites (with tertiary branches that extend into the outer molecular layer). Then, the DCX-positive cells in the granular cell layer in the DG in each section were counted. The average count from the sections of all mice was presented.

Statistical analysis

All analyses were performed using SPSS 9 software. Data are presented as mean ± SD for the total number of cells. Student's t test was performed for comparison between the Gb-treated and nontreated groups. Results were considered statistically significant when the P value was <0.05.

Western blot analysis

To confirm the changes in the DCX protein level of the DG, the DG tissues of five animals in each group were used for Western blot analysis. The tissues were homogenized in 50 mM PBS (pH 7.4) containing 0.1 mM ethylene glycol, bis (2-aminoethyl ether)-tetraacetic acid (EGTA) (pH 8.0), 0.2 % Nonidet P-40, 10 mM ethylendiamine tetraacetic acid (EDTA) (pH 8.0), 15 mM sodium pyrophosphate, 100 mM β-glycerophosphate, 50 mM NaF, 150 mM NaCl, 2 mM sodium orthovanadate, 1 mM phenylmethylsulfonyl fluoride (PMSF) and 1 mM dithiothreitol (DTT). After centrifugation, the protein level in the supernatant was determined using a Micro BCA protein assay kit with bovine serum albumin as the standard (Pierce Chemical, Rockford, IL, USA). Aliquots containing 20 μg of total protein were boiled for 5 min in loading buffer containing 150 mM Tris (pH 6.8), 3 mM DTT, 6 % SDS, 0.3 % bromophenol blue and 30 % glycerol. The aliquots were then loaded onto a 10 % polyacrylamide gel. After electrophoresis, the gels were transferred to nitrocellulose transfer membranes (Pall Corp., East Hills, NY, USA). To reduce background staining, the membranes were incubated with 5 % nonfat dry milk in PBS containing 0.1 % Tween 20 for 45 min, followed by incubation with goat anti-DCX (1:100), peroxidase-conjugated rabbit anti-goat IgG (Sigma) and an enhanced luminol-based chemiluminescent (ECL) kit (Pierce Chemical). The blot was densitometrically scanned for the quantification of the relative optical density (ROD) of each band using Scion Image software (Scion Corp., Frederick, MD, USA), which was used to count the ROD. These data were normalized with β-actin.

Results

Histological examination of DG hippocampal sections

Normal structure of the dentate gyrus (DG) of young adult male mice shows that the DG consists of three layers, an outer molecular layer (ML), principle crowded middle

granular cell layer (GCL) and inner polymorphic layer (PL). The polymorphic and molecular layers are the least cellular layers; they are formed of a few neurons and glial cells in between many neuronal processes. The GCL is formed of 5–6 rows of closely packed neurons with rounded vesicular nuclei, prominent nucleoli and a scanty basophilic cytoplasm. A few dark nuclei appear in the deep layer of the granular cells (Fig. 1a). Sections of the GCL of the aged group showed eosinophilic (degenerating) neurons with darkly stained (pyknotic) nuclei. Swollen cells and vacuolations were also demonstrated. Some dark irregular shrunken neurons were detected. There was apparent decreased cellularity in the GCL (3–4 rows) compared to young mice (Fig. 1b). The Gb-treated group revealed a picture nearly similar to that of young mice with apparently more granule cells and fewer dark neurons compared to the nontreated elderly mouse group (Fig. 1c).

To confirm the presence or absence of apoptosis in different groups, we performed immunostaining for anti-caspase-3 in the DG of the normal young adult animals, negative control (normal elderly mice) and Gb-treated groups: Hippocampal sections of the young adults revealed -ve immunoreactivity in all layers of the DG except very few positive cells scattered throughout the whole DG (Fig. 2a). The negative control (normal elderly mice) group revealed brown cytoplasmic immunostaining in the majority of DG granule cells mainly in the SGZ (Fig. 2b). In the Gb-treated group there were few cells positive for caspase-3 in the GCL of the DG (Fig. 2c).

GFAP+e stem cells in the DG are increased by Gb treatment

Hippocampal sections of the control group (Fig. 3a) revealed brown immunostaining in the cytoplasm and processes of astrocytes in regions of the DG in the molecular and polymorphic layers (red arrows), while the GFAP+ve immunoreactive cells of the granular cell layer were detected mainly in the subgranular zone (SGZ) of this layer (black arrows) (Fig. 3a). The Gb-treated group (Fig. 3b) revealed apparently increased immunoreactivity as compared to the control. Also, in the Gb-treated group, an increased number of immunoreactive astrocytes of the granular cell layer was detected mainly in the SGZ (black arrows) (Fig. 3b). Counting the immunoreactive GFAP+ve cells in the GCL of both groups (Fig. 3c) showed that there was a significant increase ($P < 0.002$) in the mean of the total number of GFAP+ve cells in the GCL of Gb-treated mice (96 cells/GCL) compared to that of the nontreated group (22 cells/GCL).

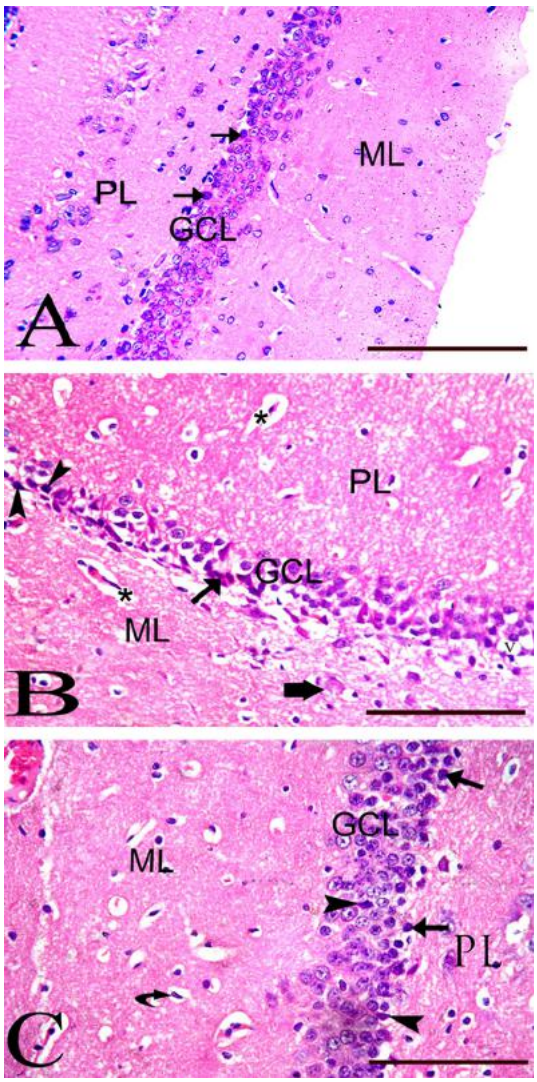

Fig. 1 Hematoxylin and eosin-stained DG hippocampal sections: **a** Photomicrograph of a section of the DG of the hippocampus of a normal young adult male mouse. The three layers, the PL, GCL and ML, are observed. The PL and ML show few neurons and glial cells in between the neuronal processes. The GCL is formed of several layers of closely packed neurons with rounded vesicular nuclei, prominent nucleoli and a scanty basophilic cytoplasm. Few dark nuclei appear in the deep layer of granular cells (*arrows*). **b** A section of the DG of the hippocampus of an albino rat from old nontreated mice showing the three layers: PL, GCL and ML. Eosinophilic (degenerating) neurons with darkly stained (pyknotic) nuclei are detected (*black arrows*). Dark irregular shrunken neurons (*arrowheads*), a swollen cell (*thick arrow*) and blood vessels (*asterisk*). **c** A section of the DG of the hippocampus of mice of the Gb-treated group: the three layers, PL, GCL and ML, are examined. The GCL has several layers of closely packed, rounded vesicular nuclei with prominent nucleoli and scanty basophilic cytoplasm. Few dark nuclei appear in the deep GCL (*arrows*). Few dark irregular shrunken cells are detected (*arrowheads*). Note the typical rod-shaped microglial nucleus (*curved arrow*). *PL* polymorphic layer, *GCL* granular cell layer, *ML* molecular cell layer; *scale bar* ×400

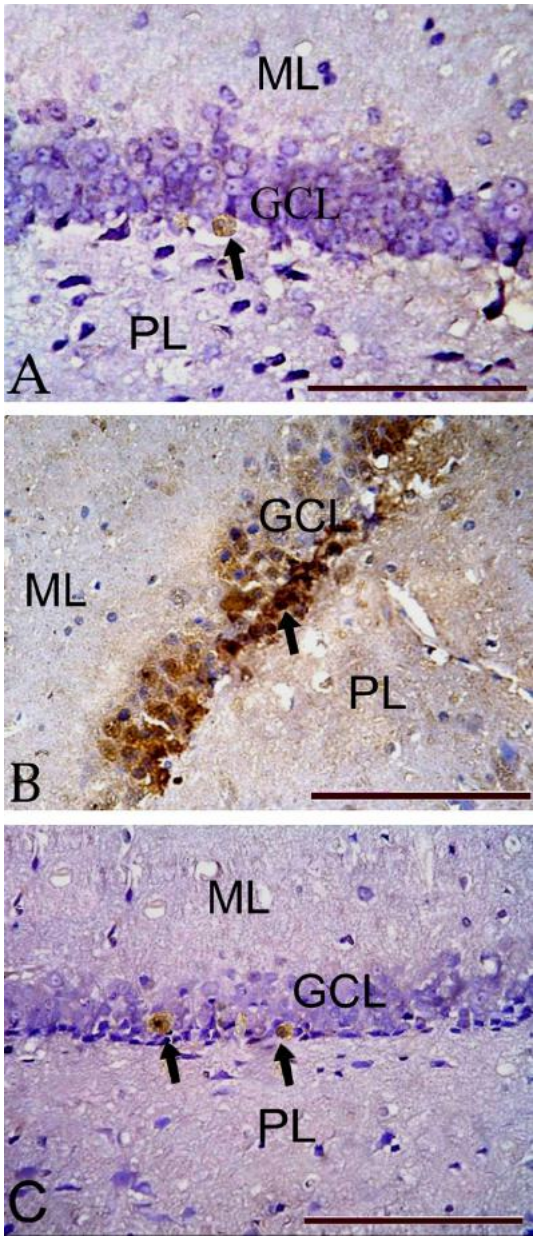

Fig. 2 Anti-caspase-3 immunostaining in the DG hippocampal sections. Photomicrographs of a section in the hippocampal DG. Very few cells show +ve immunoreactivity throughout the whole DG in young adult mice in **a**, and also few positive cells (*arrows*) are seen in Gb-treated elderly mice (**c**). In old-aged nontreated mouse group (**b**) brown cytoplasmic immunostaining is detected in most cells of the GCL (*arrow*), indicating that a large number of cells are apoptotic in this age group. *PL* polymorphic layer, *GCL* granular cell layer, *ML* molecular cell layer; *scale bar* ×400

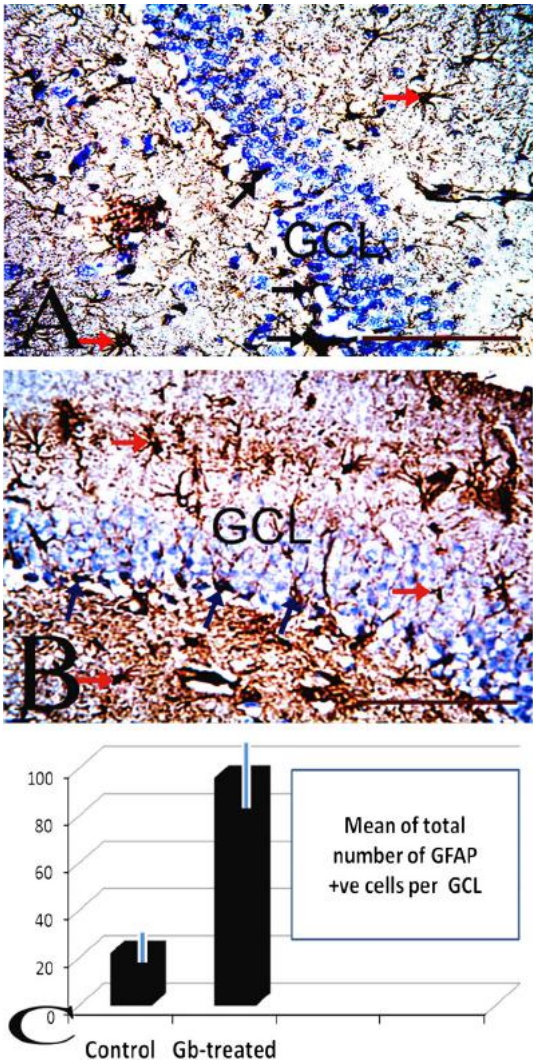

Fig. 3 Anti-GFAP immunostaining in the DG hippocampal sections: **a** Photomicrograph of a section of the DG of the hippocampus of an old mouse of the nontreated group showing brown immunostaining in the astrocyte cytoplasm and processes in the three layers of the DG, mainly in the ML and PL (*red arrows*). The immunoreactive cells of the GCL are detected mainly in the subgranular zone (*black arrows*). **b** Photomicrograph of a section of the DG of the hippocampus of the old Gb-treated mouse group showing generally increased immunoreactivity of GFAP+ve in the whole section compared to that of the non-treated group. Brown immunostaining is detected in the cytoplasm and processes of many astrocytes in all layers of the DG (*red arrows*). The immunoreactive cells of the GCL are detected mainly in the SGZ (*black arrows*). **c** Histogram of the total number of GFAP-expressing cells in the GCL of the whole DG showing that the mean total number of GFAP+ve cells is 22 cell/GCL in the nontreated group but this number increases more than fourfold in the Gb-treated mice (96/GCL). *Bars* mean ± SD; P value < 0.002. GCL, granular cell layer; *scale bar* ×400

Gb increased cell proliferation in the DG

The sections were immunostained for the endogenous proliferation marker Ki-67. There were very few immunoreactive Ki-67 cells in the control group (Fig. 4a), while the Gb-treated group revealed brown nuclear immunoreactivity that was detected in the granular cell layer of the DG mainly in the subgranular zone. Few immunoreactive nuclei were detected in the polymorphic and molecular layers in the Gb-treated

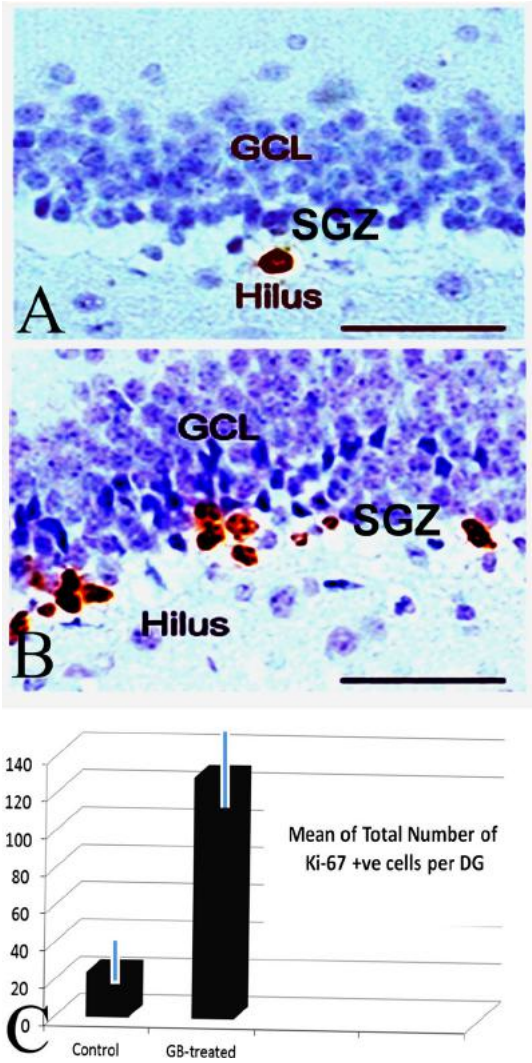

Fig. 4 Anti-Ki67 immunostaining in the GCL of DG hippocampal sections: **a** Photomicrograph of a section of the DG of the hippocampus of an old mouse of the nontreated group showing few brown immunostained cells for Ki-67 located in the nuclei of cells in the subgranular zone of the GCL of the DG. **b** In the Gb-treated group there are clusters of Ki67 brown-stained nuclei in the GCL of the DG. Note that Ki-67+ve cells are found in the subgranular zone of the DG. The histogram shows that the total number of Ki-67-positive cells in the Gb-treated group is significantly higher than that of the nontreated group. *Bars* mean ± SD; $P < 0.003$. GCL, granular cell layer; SGZ, subgranular zone; *scale bar* ×400

group (Fig. 4b). The total number of Ki-67 immunoreactive cells was counted in the granule cell layer (GCL) and the subgranular zone of the DG. The Gb-treated group showed a significantly increased number of Ki-67-expressing cells in the GCL of the DG compared to the nontreated group ($P < 0.001$) (Fig. 4c). These results indicate that the Gb increased cell proliferation in the subgranular zone of the DG of the hippocampus.

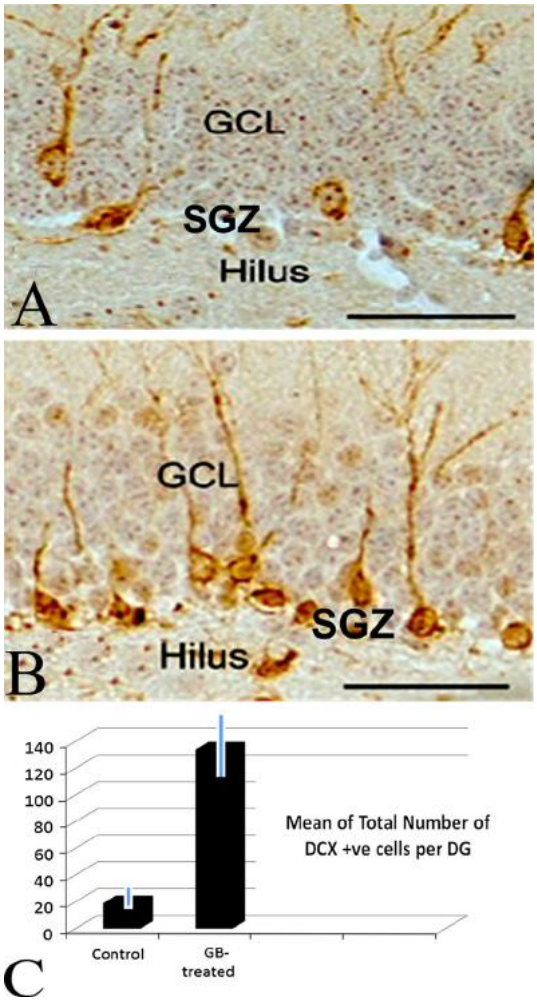

Fig. 5 Anti-DCX immunostaining in the GCL of DG hippocampal sections: **a** Photomicrograph of a section of the hippocampus of an old mouse of the nontreated group shows brown immunostained neurons for DCX in the GCL of the DG. The Gb-treated group showed more cells that were stained for DCX in the GCL (**b**). Note that DCX+ve cells are found in the subgranular zone of the DG. The histogram shows that the total number of DCX-positive cells in the Gb-treated group is higher than that in the nontreated group. *Bars* mean ± SD; $P < 0.001$. Note that in the Gb-treated group the DCX-immunoreactive neuroblasts have well-developed processes that extend to the molecular layer of the DG. This type of DCX-positive cell with tertiary dendrites is more mature than that with simple dendrites. GCL, granular cell layer; SGZ, subgranular zone. *Scale bar* ×400

Gb increased the number and maturity of newborn neurons

We next examined the effect of Gb on newborn neurons, identified by the expression of the neuronal marker, doublecortin (DCX), for identification of migrating newborn neurons and their maturation. In the nontreated group few cells were found to be immunostained for DCX and found

in the subgranular zone of the GCL of the DG (Fig. 5a). With Gb treatment, there was an increase in the total number of DCX-immunostained neuronal soma in the subgranular zone of the GCL (Fig. 5b, $p < 0.002$). Furthermore, upon Gb treatment, migrating neurons displayed an increased proportion of dendritic branches extending into the molecular layer, suggesting increased maturation (Fig. 5b). To further investigate the effect of Gb on the new neurons' maturation, we quantified three categories of DCX-expressing cells (Seri et al. 2004): cells without processes, known to be the most immature subtype with the highest proliferative activity; cells with few and horizontal processes; and cells with tertiary and vertical processes extending to the GCL, which represent the most mature phenotype (Fig. 5b). The GB significantly increased the proportion of DCX-expressing cells with tertiary processes, i.e., an increased number of the most mature type of newborn neurons ($P < 0.001$). All together, these results indicated that Gb treatment increased the number of new migrating neurons and increased the proportion of cells with mature morphological characteristics. Western blot analysis revealed that DCX protein was significantly increased compared to the control group (Fig. 6) (P value < 0.05).

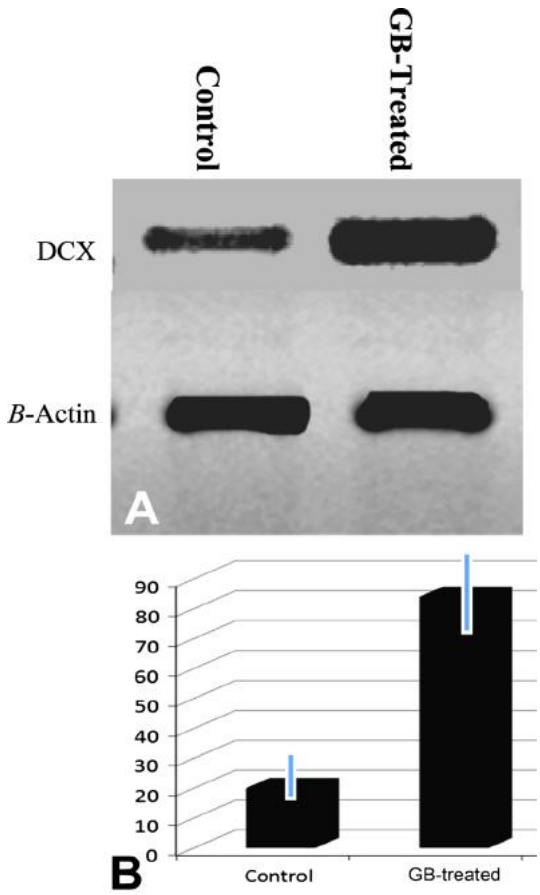

Fig. 6 Western blot analysis of DCX protein in the DG hippocampal tissues: the density of the immunoblot bands of DCX protein indicates that the Gb-treated group has a higher content of DCX protein than the nontreated one in the DG. As shown in the blot in **a** and the histogram in **b** for the relative optical density (ROD). The ROD of the immunoblot band of the GB-treated group is indicated as percent values versus the control group ($n = 5$ per group; $P < 0.05$). *Bars* mean ± SD

Discussion

Neurogenesis declines with age in the two neurogenic regions of the adult brain, the dentate gyri of the hippocampus (Seki and Arai 1995; Kuhn et al. 1996; Kempermann et al. 1998; Cameron and McKay 1999; Dupret et al. 2007) and the subventricular zone of the lateral ventricles (Tropepe et al. 1997; Jin et al. 2003). However, some capacity for neurogenesis is maintained in old age. The finding that the intraventricular infusion of growth factors in elderly mice can stimulate an increase in neurogenesis back to levels found in the young adult (Jin et al. 2003) suggests that the basic components of neurogenesis can be retained by different factors (Kheirbek 2015).

In the DG, new neurons originate from cells located in the subgranular layer (SGL) at the border of the granule cell layer (GCL) facing the hilus. These cells are slowly proliferating astrocytes [identified on the basis of their expression of glial fibrillary acidic protein (GFAP) and their ultrastructural properties], presumably the radial glia (Seri et al. 2004). The early neurogenesis response in the dentate gyrus is associated with an increase in the divisions of radial glial cells in the subgranular zone (Hüttmann et al. 2003), which show very limited or no proliferative activity under normal conditions (Kronenberg et al. 2003; Seri et al. 2004; Suh et al. 2007). The majority of the GFAP+ve hilar proliferating cells or tertiary germinal cells were found to be neural progenitors that produce neuroblasts in the dentate gyrus (Namba et al. 2009; Liu et al. 2010). In our study there was a three times increased number of GFAP+ve cells in the subgranular zone in the Gb-treated group compared to the nontreated elderly mice, indicating that the pool of stem cells is increased and thus the proliferative capacity of the DG is increased by Gb treatment.

Cell proliferation can be studied using an intrinsic proliferation marker, Ki-67, which is transiently expressed by cycling cells. It is considered to be a reliable marker to assay the proliferative activity of precursors (Kee et al. 2002; Wojtowicz and Kee 2006; Taupin 2007). Ki-67 labels cells in G1, S, G2 and mitosis (Scholzen and Gerdes 2000). Many studies have found an overall age-related decline in cell proliferation in the brain (Kuhn et al. 1996; Kempermann et al. 1998; Cameron and McKay 1999; Lemaire et al. 2006; Molofsky et al. 2006). These previous

results were consistent with those found in our study that showed few Ki-67 immunoreactive cells, indicating a decline in the proliferative activity of precursor cells in the DG of nontreated elderly mice, while the increase in Ki-67-immunoreactive cells in the DG of Gb-treated elderly mice in this present study is supported by a pervious study that reported that 100 mg/kg Gb treatment for 1 month significantly enhanced cell proliferation in the DG of both adult (6 months) and aged (22 months) Alzheimer mice compared with their untreated counterparts (Shif et al. 2006).

Doublecortin (DCX) is a protein that promotes microtubule polymerization and is expressed in migrating neuroblasts and young neurons (Von Bohlen und Halbach 2007). It was found that GFAP-expressing radial-like cells in the DG are the precursor cells that divide asymmetrically to generate a daughter radial cell and a DCX-expressing direct progenitor, which is a neuroblast or immature neuron (Seri et al. 2004; Encinas et al. 2006). In our study, the DCX-immunoreactive cells were few in number in the nontreated elderly mice. This observation is similar to what is observed with cumulative labeling studies in which the number of DCX-labeled cells in the granular cell layer (GCL) of the DG is also dramatically decreased with age and becomes very low in both middle-aged and old animals (Alonso 2001; Nacher et al. 2003).

Previous reports showed that the DCX-immunoreactive neurons that exhibit primary or basal dendrites are characteristic of immature neuroblasts in rodents (Jones et al. 2003; Rai et al. 2007). In contrast, neurons with complex and vertically oriented dendrites (which extend into the outer molecular layer) are the mature ones among the population of DCX+ve neurons in the DG (Rao and Shetty 2004; Rao et al. 2005). Therefore, in our study based on immunohistochemistry and Western blot analysis, we found that the administration of Gb in elderly mice significantly increased both the DCX protein and number of DCX immunoreactive neurons with tertiary dendrites. This indicates an increase in cell differentiation and maturation of newborn neurons and neuroblasts in the hippocampus by Gb treatment in old-aged mice. These data are in agreement with previous observations that repeated treatments with Gb for aged rats elicited robust long-term hippocampal potentiation (Williams et al. 2004; Wang et al. 2006). In addition, Gb treatment shows dendritic aberrations and well-developed dendritic branches in the hippocampal progenitor cells derived from rats in vitro (Tchantchou et al. 2009).

This study detected enhanced neurogenesis with Gb-treatment in elderly mice using immunohistochemical markers to detect proliferation of cells and formation and differentiation of new neurons in the hippocampus. However, we do not know whether the increased GFAP+ve cells in this study were the source of these newborn neurons or not. What is the effect of Gb-enhanced neurogenesis on the hippocampal function in elderly mice? These questions need further investigation.

Conclusion

The results show that Gb leaf extract enhances adult neurogenesis and support a beneficial role of Gb leaf extract in the decline of hippocampal neurogenesis that accompanies brain aging.

Compliance with ethical standards

Conflict of interest None.

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